Logistic Regression Interpretable Baseline

Preprocessing

1) **Data loading:** we used the “master_dataset_transformed_after_1986.csv”.

2) **Data Cleaning (Handling Unreliable Features: "TEDRATE"):**

   There are 3 empty columns that we dropped ["BAA10Y_missing", "T10Y2Y_missing", “FEDFUNDS_missing"], they didn't contribute anything to the model but nois. Moreover, after carefully validating the sources we found that in 2022 the fed stoped measuring the “Tedrate” index, due to manipulative behaviour (“Libor scandal”, “Does the LIBOR Reflect Banks' Borrowing Costs?”).

3) **Data Engineering:**

   Before modeling, the dataset was rigorously prepared to ensure economic reality and prevent look-ahead bias:

   Lag Construction: Shifted variables were created for t-1, t-2, and t-3 quarters to capture the delayed impact of macroeconomic shifts.

   Rolling Statistics: Rolling averages corresponding to 3M, 6M, and 12M (represented as 1-quarter, 2-quarter, and 4-quarter rolling means in the quarterly dataset) were engineered to smooth high-frequency noise and capture macroeconomic momentum.

   Regime Indicators: A pre/post-2008 (Subprime Crisis) and 2020 (Covid-19) dummy variable was included to allow the model's intercept to adjust for the structural shift in the economic and monetary environment following the Global Financial Crisis.

   Strict Anti-Leakage Checks: Target variables (Target_1Q_ahead, Target_2Q_ahead, Target_3Q_ahead) were carefully shifted backwards relative to the features. Crucially, all imputation (SimpleImputer) and scaling (StandardScaler) were strictly fitted only on the training windows and then transformed on the test windows within the cross-validation loop.

Logistic Regression Baseline Modelling

1) Introduction & Model Setup

To establish a robust and interpretable foundation for this analysis, Logistic Regression models were built for three distinct macroeconomic forecasting horizons: 1Q, 2Q, and 3Q ahead.

Because recessions are relatively rare macroeconomic events, standard classification algorithms tend to overpredict the majority class (economic expansion). To counter this imbalance, class weighting (class_weight="balanced") was applied across all models. This forces the loss function to heavily penalize misclassifications of recessionary periods. By utilizing logistic regression, the resulting coefficients remain economically interpretable, directly representing the change in the log-odds of a recession for a one-unit change in the underlying macroeconomic indicator.

2) Feature Selection (Mutual Information + LASSO)

The feature engineering phase generated a high-dimensional dataset prone to severe multicollinearity. To distill this into a clean signal, a two-step feature selection process was implemented:

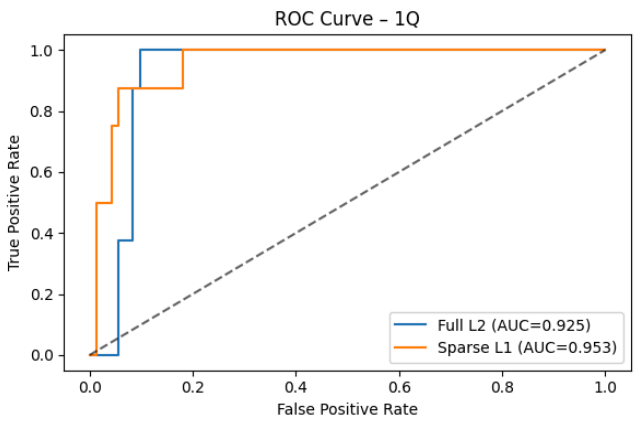
Mutual Information: Features were initially ranked using Mutual Information (mutual_info_classif) to identify non-linear dependencies between the predictors and the recession target, establishing a descriptive baseline of feature importance.

LASSO Regularization: An L_1-penalized logistic regression (sparse_l1 with C=0.25) was implemented to enforce sparsity. The LASSO penalty successfully drove the coefficients of redundant variables to zero, effectively performing algorithmic feature selection.

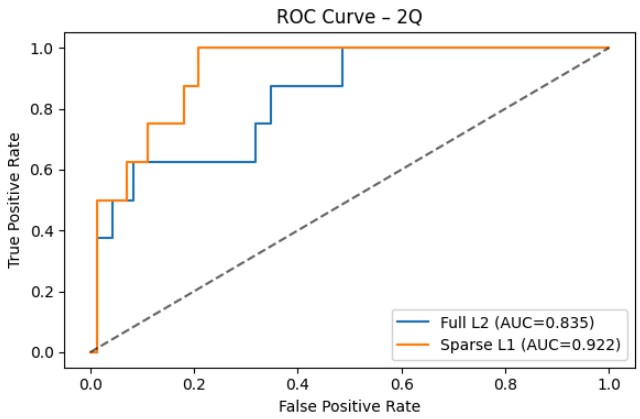
3) Model Comparison:

When compared against the standard L_2-penalized model (full_l2), the sparse L_1 model provided comparable predictive power but with a fraction of the features, drastically reducing overfitting and making the final economic narrative much easier to interpret.

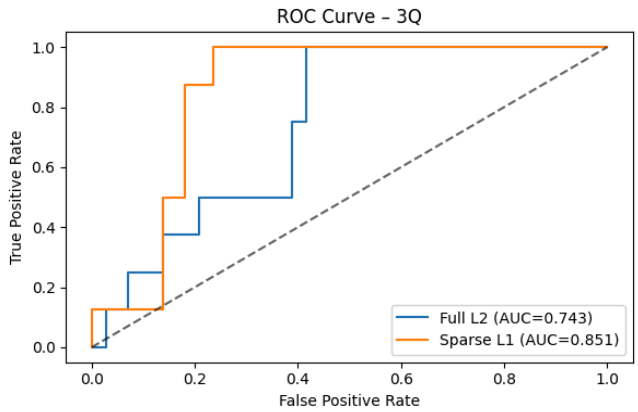
1Q Horizon (One quarter ahead): The full model achieved an AUC of 0.925 compared to 0.953 for the sparse model. This represents a negligible improvement of 3.02%.



2Q Horizon (Two quarters ahead): The full model achieved an AUC of 0.835 compared to 0.922 for the sparse model, representing a moderate improvement of 10.4%.



3Q Horizon (Three quarters ahead): The full model achieved an AUC of 0.743 compared to 0.851 for the sparse model, representing a more significant improvement of 14.5%.



In all of the horizon periods, the sparse model is highly effective and completely sufficient; it achieves better performance while utilizing significantly fewer variables.

4) Training & Validation Framework

Standard random k-fold cross-validation is invalid for time-series data as it leaks future information into the past. To evaluate the models accurately, a strict time-aware backtesting strategy was required.

Expanding vs. Rolling Window Validation: An expanding window approach (expanding_splits) was selected as the preferred method. The model was initially trained on an anchored history of 60 quarters, and tested on an out-of-sample block of 8 quarters. The training window then expanded step-by-step. Because recessions are rare events, retaining the full history ensures the model does not "forget" past recession patterns. A rolling

window, while useful for adapting to structural regime shifts, risks discarding critical minority-class examples once they fall outside the training window.

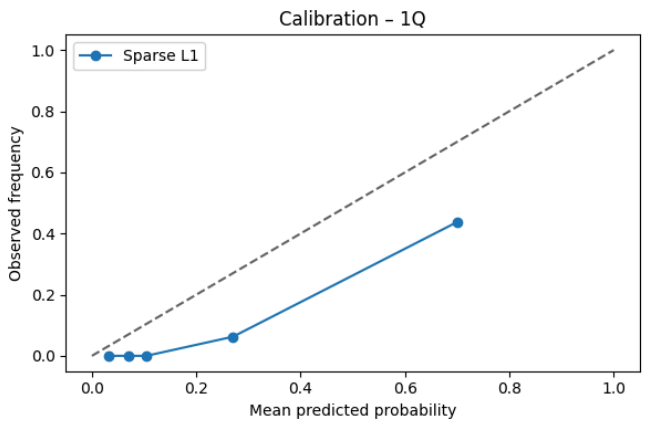
Target Autocorrelation & Model Statelessness: Standard algorithms like Logistic Regression are inherently stateless; they evaluate each time step (t) independently without explicitly learning temporal sequences. The model outputs consecutive recession predictions not because it "remembers" the previous quarter, but because the macroeconomic input features (X) are highly autocorrelated. The persistence of poor economic conditions naturally leads to persistent predictions.

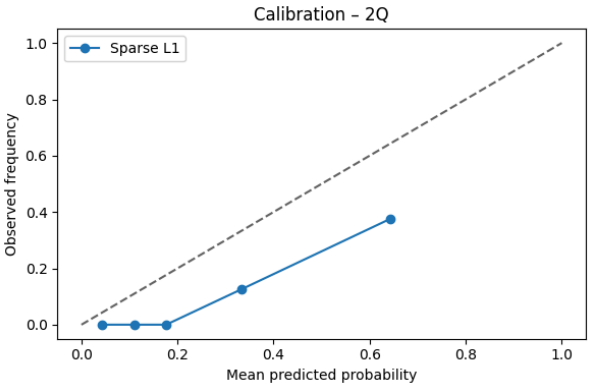
Multi-Quarter Horizons as a Natural Embargo: Predicting extended horizons (2Q or 3Q ahead) acts as a built-in "embargo" mechanism. By shifting the target variable forward, a natural temporal gap is created between the observed features and the predicted event. This buffer helps mitigate immediate serial correlation and prevents short-term data leakage, forcing the model to rely on genuine leading indicators rather than overlapping, short-term macroeconomic momentum.

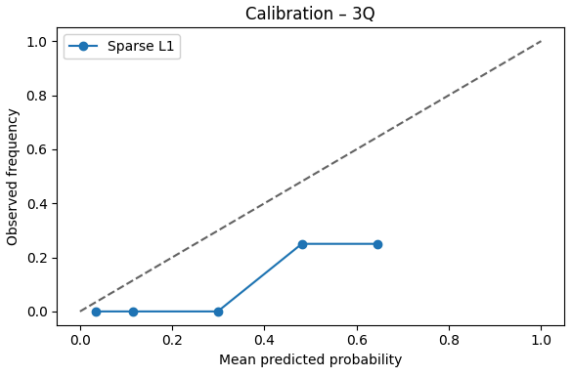
5) Diagnostics & Statistical Inference

Raw classification metrics can be misleading if the model's output probabilities do not reflect reality. To ensure statistical rigor, the pipeline incorporated both calibration and significance testing.

Calibration: Probability calibration curves (Reliability Diagrams) were plotted.







A cross-horizon comparison of the reliability diagrams reveals a clear trend of calibration decay as the forecasting window extends. Across all three horizons, the models systematically over-forecast the absolute probability of a recession (the blue lines sit below the dashed diagonal). **As established, this is the intended mathematical consequence of applying class_weight="balanced" to heavily penalize false negatives in a highly imbalanced dataset.**

However, the severity of this over-forecasting worsens significantly from 1Q to 3Q.

- **1Q Horizon:** The model is best calibrated. A mean predicted probability of ~70% aligns with an observed historical frequency of ~40%.
- **2Q Horizon:** The calibration begins to flatten, with high-probability predictions (~65%) mapping to an observed frequency of roughly 38%.
- **3Q Horizon:** The calibration degrades severely. When the model issues a stark warning (predicting a ~65% probability of recession), a recession only materialized in reality about 25% of the time.

Significance Testing: To demonstrate strict statistical discipline, the sparse features selected by the LASSO model were passed into an econometric model (statsmodels.Logit). This allowed for the generation of descriptive coefficient tables containing p-values, standard errors, and 95% confidence intervals. This proves that the selected features have statistically significant predictive power ($p < 0.05$), moving the analysis beyond "black-box" predictive scoring and into robust economic inference.

6) Horizon Comparison

The pipeline explicitly tracked out-of-sample performance metrics (AUC, Average Precision, Brier Score, and Log Loss) across the 1Q, 2Q, and 3Q horizons.

As expected, the models exhibited predictive degradation as the forecasting horizon extended. The 1Q ahead model performed the best, 2Q remained highly useful, and 3Q showed noticeable decay in precision and AUC.